Football Metrics Consolidation & Ontology Encyclopedia

The Landscape of Football Metrics Platforms

- Tiers of Data Providers
  - Premium providers offer sophisticated metrics with proprietary machine learning models
  - Mid-tier platforms provide comprehensive coverage with simpler calculations for scouting
  - Public platforms democratize access to advanced metrics with transparent methodologies
- Leading Platforms and Innovations
  - StatsBomb leads with 360 freeze-frame data and On-Ball Value (OBV) model
  - Opta/Stats Perform dominates market share, powering FBref and broadcasters
  - Wyscout excels in coverage breadth with 600+ leagues and unique metrics like Smart Pass

Canonical Metric Families: The Unified Taxonomy

- Expected Goals (xG) Family
  - Quantifies shot quality by estimating probability of a goal based on historical shots
  - StatsBomb uniquely includes freeze-frame data (defender positions, goalkeeper location)
  - Wyscout's model assigns systematically higher xG values with a heavier tail distribution
  - Opta and Understat models correlate highly, especially over season aggregates
- Post-Shot Expected Goals (PSxG) Family
  - Evaluates shot quality after the shot is taken, measuring placement difficulty for goalkeepers
  - PSxG - xG = Shooting Goals Added (SGA), isolating pure finishing skill
  - Elite finishers consistently show positive SGA, indicating superior shot placement
- Possession Value Family
  - Values every on-ball action based on its contribution to scoring and preventing goals
  - xT is location-based; VAEP is feature-rich; OBV includes goals-against component
  - g+ uniquely splits pass credit between passer and receiver
  - EPV requires tracking data and evaluates off-ball positioning
- Pressing Intensity Family
  - Quantifies how intensely a team presses the opposition
  - PPDA is opponent passes / defensive actions in attacking 60% of pitch
  - StatsBomb's Pressure Events define pressure as player within 5 yards of ball carrier
  - PPDA can conflate pressing with possession dominance, showing artifically low values for high-possession teams
- Progressive Passing Family
  - Measures passes that significantly move the ball closer to the opponent's goal
  - Definitions vary by platform regarding distance thresholds and pitch zones
  - FBref excludes passes from defensive 40%, Wyscout uses tiered distance thresholds
- Shot-Creating Actions Family
  - Measures offensive actions directly leading to a shot or a goal
  - SCA/GCA (FBref/StatsBomb) are two actions leading to a shot/goal
  - Opta's Key Pass is the final pass to a teammate who attempts a shot
  - SCA components include passes, dribbles, shots (rebounds), fouls drawn, and defensive actions
- Deep Penetration & Smart Pass Families (Wyscout-specific)
  - Deep Completion: Non-cross pass into zone within 20m of opponent goal
  - Deep Completed Cross: Cross into zone within 20m of opponent goal
  - Smart Pass: Creative, penetrating pass attempting to break defensive lines, identified by human taggers
  - These Wyscout metrics have no direct equivalents on other platforms
- Aerial Ability Family
  - Measures success in aerial duels
  - StatsBomb HOPS is a Glicko rating system accounting for opponent strength
  - Height explains only ~22% of aerial duel variance; technique, timing, and aggression are more crucial
- Player Rating Family
  - Provides an overall performance index for players
  - InStat and Wyscout use position specific parameters and weights
  - SofaScore uses a universal ML algorithm with position-adjusted weights, weighting offensive actions higher

Physical and Tracking Metrics

- Distance and Speed Family
  - Includes Total Distance, High Speed Running (HSR), Sprinting, Accelerations, and Metabolic Power
  - Optical tracking underestimates accelerations by ~47% compared to GPS-MA devices
- Off-Ball Runs Family (SkillCorner)
  - Categorizes player movements without the ball, such as Runs in Behind and Cross Receiver Runs
  - Metrics are normalized per 30 minutes Team In Possession (P30 TIP)
- Pitch Control Family
  - Probability surface showing likelihood of each team controlling any point on pitch
  - Based on player positions, velocities, and time-to-intercept

HP Engine Framework Mapping

- Metric Distribution Across Categories
  - Pre-Match Analysis uses PPDA, formation metrics, historical xG trends
  - Post-Match Analysis uses xG, PSxG, VAEP, OBV, GCA, physical output
  - Individual Analysis uses possession value, SCA, xA, player ratings, HOPS
  - Team Tactical Analysis uses PPDA, Pitch Control, build-up attacks, pass networks
- Metric Mapping to Game Phases
  - Defensive Organization uses PPDA, recoveries, interceptions, aerial duels
  - Defense → Attack Transition uses Counter Press, high turnovers
  - Build-Up uses xSuballdup, Progressive Passes, xT from carries
  - Attacking Organization uses xG, xA, SCA/GCA, Deep Completions
  - Attack → Defense Transition uses PPDA, Pressure Events, ball recoveries
- Analytical Layer Classification
  - Micro layer: Individual action (xG per shot, VAEP per action)
  - Mezzo layer: Sequence/pattern (xGChain, Build-up Attacks)
  - Macro layer: Team/match (Total xG, PPDA, xPTS)

Platform Translation Table

- Canonical to Platform Mapping
  - Provides a cross-platform reference for key metric families
  - Shows how Expected Goals, Pressing Intensity, Progressive Passes, and others are named across platforms

Metric Combinations Guide (Metrik Birleştirme Kılavuzu)

- Finishing Skill Analysis
  - Combines PSxG and xG to calculate Shooting Goals Added (SGA = PSxG - xG)
  - Answers if a forward is a technically superior finisher or good at finding positions
- Goalkeeper Evaluation
  - Combines PSxG Faced and Goals Conceded to calculate PSxG+/-
  - Evaluates goalkeeper shot stopping ability independent of team defense
- Pressing Effectiveness
  - Combines PPDA, Buildup Disruption Percentage (BDP), and Recovery Height
  - Determines if pressing is efficient and intelligent, not just intense
- Ball Progression Analysis
  - Combines Progressive Passes, xT from Passes, and Carries into Final Third
  - Assesses how effectively a player carries the ball into dangerous areas
- Complete Attacking Output
  - Combines np:xG and xA (per 90) to get np:g+/xA
  - Provides a single metric for a player's total expected offensive contribution
- Creative Quality vs Output
  - Combines xA (Expected Assists) and Actual Assists to calculate Creative Differential
  - Determines if a playmaker's assist count reflects true creativity or fortunate quality

JSON Structure (Sample Canonical Families)

- Expected Goals (xG)
  - Details purpose, core question, platform calculations, data requirements, and combinations
  - Example combination: PSxG - xG produces Shooting Goals Added (SGA)
- Possession Value (OBV/VAEP)
  - Details purpose, core question, platform methodologies, and data requirements
  - Example combination: Sum of xT/VAEP from progressive actions produces Progression Value
- Pressing Intensity
  - Details purpose, core question, platform metrics, and benchmark values
  - Example combination: BDP / inverse(PPDA) produces Pressing Efficiency
- Shot-Creating Actions
  - Details purpose, core question, platform definitions, and benchmark values
  - Example combination: SCA weighted by xA produces Creative Quality Score
- Deep Completion (Wyscout)
  - Details purpose, core question, platform definitions, and benchmark values
  - Example combination: xS from deep completions / Deep completions produces Deep Penetration Efficiency

Conclusion: A Unified Framework for Turkish Football Analytics

- Consolidation Principles
  - Identical metrics across platforms are single canonical entries
  - Different names, identical purpose are merged under canonical names
  - Conceptually similar but different measurements are grouped in families but kept separate
  - Platform-specific unique metrics are documented with 'No Direct Equivalent'
  - Tracking-only metrics are in a separate section noting data requirements
- Actionable Metrics for Turkish Süper Lig
  - xG/xA, PPDA, Progressive Passes, and SCA/GCA are most actionable
  - Advanced metrics like OBV and VAEP offer comprehensive player evaluation with premium data
- Systematic Metric Selection with HP Engine
  - Macro-layer metrics for team tactical analysis
  - Mezzo-layer metrics for pattern identification
  - Micro-layer metrics for player scouting
  - Combining metrics across layers yields the deepest tactical insights